

Swiggy Food Delivery Time Prediction Using Sagemaker

In [13]:

```
import boto3
import pandas as pd
import joblib
import sklearn
import sys

import warnings
warnings.filterwarnings("ignore")

s3 = boto3.client("s3")

bucket = "swiggydeliverytimepredmlops-347794108556-ap-southeast-2-an"

dataset_key = "dataset/swiggy.csv"
model_key = "ml_model/swiggy_delivery_time_pred_model.pkl"

s3.download_file(
    bucket,
    dataset_key,
    "swiggy.csv"
)

print("Dataset downloaded successfully!")

df = pd.read_csv("swiggy.csv")

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)

display(df.head())

s3.download_file(
    bucket,
    model_key,
    "Swiggy_FoodDeliveryTimePrediction_Model.pkl"
)

print("Model downloaded successfully!")

model = joblib.load(
    "Swiggy_FoodDeliveryTimePrediction_Model.pkl"
)

print("Model loaded successfully!")
print("Model type:", type(model))

single_data = pd.DataFrame({
    "Area": ["Kanyakumari"],
    "City": ["Bangalore"],
    "Restaurant": ["Kim Lee"],
```

```

    "Price": [50.0],
    "Avg ratings": [4.4],
    "Address": ["Double Road"],
    "Total ratings": [400]
})

# =====
# PREDICTION
# =====

pred = model.predict(single_data)

print(
    "Predicted Delivery Time:",
    round(pred[0]),
    "minutes"
)

```

Dataset downloaded successfully!
Dataset loaded successfully!
Dataset shape: (8680, 10)

	ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address	Delivery time
0	211	Koramangala	Bangalore	Tandoor Hut	300.0	4.4	100	Biryani,Chinese,North Indian,South Indian	5Th Block	59
1	221	Koramangala	Bangalore	Tunday Kababi	300.0	4.1	100	Mughlai,Lucknowi	5Th Block	56
2	246	Jogupalya	Bangalore	Kim Lee	650.0	4.4	100	Chinese	Double Road	50
3	248	Indiranagar	Bangalore	New Punjabi Hotel	250.0	3.9	500	North Indian,Punjabi,Tandoor,Chinese	80 Feet Road	57
4	249	Indiranagar	Bangalore	Nh8	350.0	4.0	50	Rajasthani,Gujarati,North Indian,Snacks,Desser...	80 Feet Road	63

Model downloaded successfully!
Model loaded successfully!
Model type: <class 'sklearn.model_selection._search.GridSearchCV'>
Predicted Delivery Time: 46 minutes

In []: